

```
1 using CustomProjectRPG.Interfaces;
2 using System;
3 using System.Collections.Generic;
4 using System.Linq;
5 using System.Text;
6 using System.Text.Json;
7 using System.Threading.Tasks;
8
9 namespace CustomProjectRPG.Entities
10 {
11     public class Item: IIdentifiable, ISerializeJSON<Item>
12     {
13         private string _id;
14         private string _name;
15         private string _description;
16         private bool _isUsable;
17
18         public Item(string id, string name, string description, bool
19             isUsable)
20         {
21             _id = id;
22             _name = name;
23             _description = description;
24             _isUsable = isUsable;
25         }
26
27         public string Id { get; }
28
29         public string Name { get; }
30         public string Description { get; }
31
32         public bool IsUsable { get; }
33
34         public void ApplyEffect(PlayerEntity player)
35         {
36             // Apply healing, buffs, etc.
37         }
38
39         public string Serialize()
40         {
41             var data = new
42             {
43                 Id = _id,
44                 Name = _name,
45                 Description = _description,
46                 IsUsable = _isUsable
47             };
48             return JsonSerializer.Serialize(data);
```

```
49     }
50
51     public void Deserialize(string json)
52     {
53         var data = JsonSerializer.Deserialize<Dictionary<string,      ↗
54             object>>(json) ??
55             throw new ArgumentException("Invalid JSON format.", ↗
56                 nameof(json));
57         _id = data["Id"]?.ToString() ?? throw new ArgumentException("  ↗
58             Id is required.", nameof(json));
59         _name = data["Name"]?.ToString() ?? throw new ArgumentException ↗
60             ("Name is required.", nameof(json));
61         _description = data["Description"]?.ToString() ?? throw new ↗
62             ArgumentException("Description is required.", nameof(json));
63         _isUsable = data["IsUsable"] is bool usable ? usable : throw ↗
64             new ArgumentException("IsUsable must be a boolean.", nameof ↗
65                 (json));
66     }
67 }
```